Chapter test

35 Hormonal control of the reproductive cycle

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following statements about follicles are correct?

   (1) In each menstrual cycle, FSH stimulates one follicle to develop.
   (2) The developing follicle in the ovary secretes oestrogen.
   (3) The follicle ruptures when FSH and LH burst to their peak levels.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

2. Which of the following statements about ovulation are ***incorrect***?

   (1) Ovulation is triggered by the peak levels of oestrogen.
   (2) At ovulation, the yellow body releases an ovum to the oviduct.
   (3) After ovulation, the yellow body releases oestrogen and progesterone.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

3. During a menstrual cycle, the levels of oestrogen, progesterone, FSH and LH reach its maximum level at different stages. Assuming menstruation as the beginning of the cycle, in what sequence do the hormones reach their maximum level?

   A. LH, progesterone, FSH, oestrogen
   B. FSH, oestrogen, LH, progesterone
   C. LH, oestrogen, FSH, progesterone
   D. FSH, progesterone, LH, oestrogen ☐

© 2019 Aristo Educational Press Ltd.

4. Which of the following statements about the yellow body are correct?

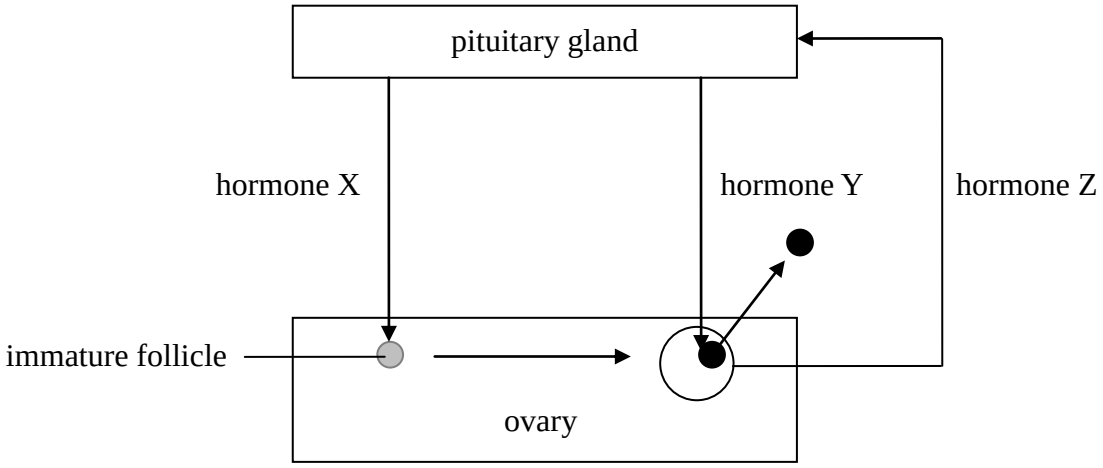
(1) It is formed from a ruptured follicle.
(2) It secretes oestrogen and progesterone.
(3) It degenerates as a result of rising levels of FSH.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

5. During the menstrual cycle, a low level of progesterone in the blood will lead to

A. thickening of the uterine lining.
B. growth of the follicle.
C. ovulation.
D. menstruation. ☐

★ 6. The diagram below shows the interactions of different hormones in the female reproductive cycle.



What are hormones X, Y and Z?

	Hormone X	***Hormone Y***	***Hormone Z***
A.	FSH	LH	oestrogen
B.	LH	oestrogen	FSH
C.	FSH	oestrogen	LH
D.	LH	FSH	oestrogen ☐

© 2019 Aristo Educational Press Ltd.

7. What is the role of human chorionic gonadotrophin in early pregnancy?

A. It prevents the degeneration of the yellow body.
B. It stimulates the degeneration of the yellow body.
C. It stimulates the thickening of the uterine lining.
D. It inhibits the secretion of oestrogen. ☐

8. Which of the following correctly explains why menstruation does ***not*** occur during pregnancy?

A. The yellow body degenerated.
B. The FSH and LH levels in a pregnant woman's blood are high.
C. The placenta secretes oestrogen and progesterone which maintain the thickness of the uterine lining.
D. Human chorionic gonadotrophin is present in a pregnant woman's blood. ☐

9. Some contraceptive pills contain a high dose of oestrogen. What is the working principle of these contraceptive pills?

A. They promote the thickening of the uterine lining.
B. They prevent the degeneration of yellow body.
C. They cause the breakdown of the uterine lining.
D. They inhibit the secretion of FSH. ☐

★ 10. Mr and Mrs W failed to keep viable pregnancy. They consulted a doctor who prescribed fertility drugs containing FSH and LH to Mrs W. How would these drugs help overcome the infertility problem?

(1) They can stimulate follicle development and ovulation.
(2) They can prevent the blockage of the oviducts.
(3) They can maintain the yellow body.

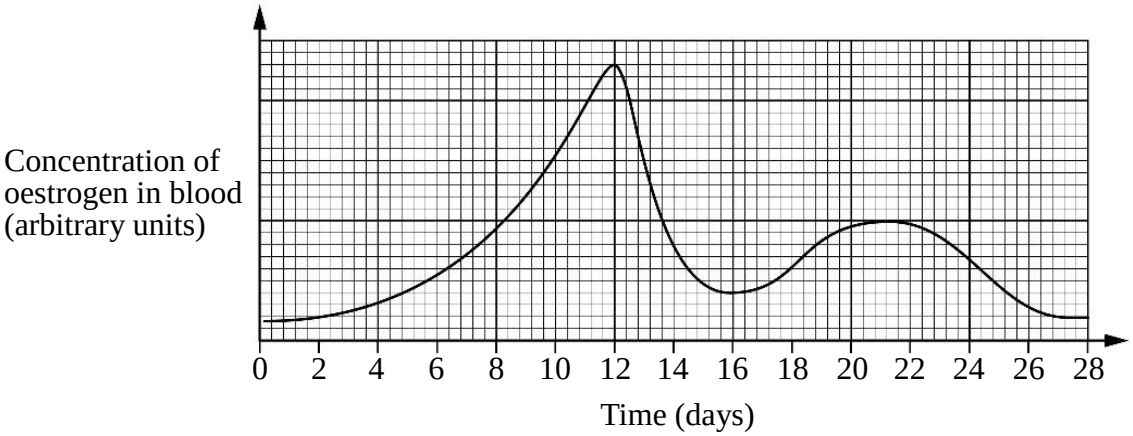
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	35.1	35.1	35.1	35.1	35.1	35.1	35.1	35.1	35.2	35.3

© 2019 Aristo Educational Press Ltd.

B. Problem solving (19 marks)

★ 11. The graph below shows the changes in the concentration of oestrogen in a woman's blood during a menstrual cycle.



(a) Name the structure that secretes oestrogen

(i) from day 1 to day 12, and (1 mark)

(ii) from day 16 to day 21. (1 mark)

(b) The concentration of oestrogen in the blood exerts different effects on the FSH and LH levels in the blood. Describe the effects of oestrogen during the following periods:

(i) from day 1 to day 9, and (2 marks)

(ii) from day 10 to day 12. (2 marks)

© 2019 Aristo Educational Press Ltd.

(c) Account for the falling concentration of oestrogen from day 22 to day 28. (4 marks)

(Total: 10 marks)

★ 12. Morning–after pills contain synthetic progesterone called levonorgestrel. Levonorgestrel works in a similar way to progesterone. It can be used to prevent or delay ovulation.

(a) Explain how levonorgestrel prevents or delays ovulation. (4 marks)

(b) Suggest ***one*** reason why morning–after pills are not 100% effective. (1 mark)

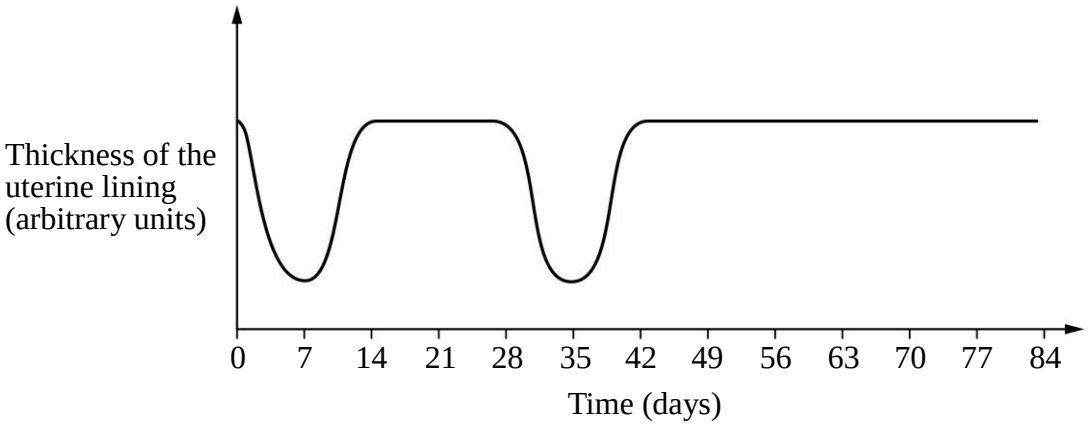
(c) Drug X blocks the action of progesterone. It binds with progesterone receptors, preventing progesterone from triggering the subsequent actions. In low doses, drug X can be used as an emergency contraceptive. In higher doses, it can be used to cause an abortion in early pregnancy. Suggest how this drug may cause abortion. (4 marks)

(Total: 9 marks)

© 2019 Aristo Educational Press Ltd.

C. Exam challenge (11 marks)

★★ 13. The graph below shows the changes in the thickness of the uterine lining of a woman over 84 days.



(a) Name the process that is taking place from day 1 to day 7. With reference to the hormones involved, account for the change in the thickness of the uterine lining during this period. (3 marks)

(b) Name the hormone(s) which

(i) cause(s) the uterine lining to thicken from day 7 to day 14. (1 mark)

(ii) maintain(s) the thick uterine lining from day 14 to day 28. (2 marks)

© 2019 Aristo Educational Press Ltd.

(c) Explain how the thickness of the uterine lining is maintained from day 56 to day 84.

(5 marks)

(Total: 11 marks)

© 2019 Aristo Educational Press Ltd.

Marking scheme

A. Concept checking (10 marks)

1.	C	2.	A	3.	B	4.	A	5.	D
6.	A	7.	A	8.	C	9.	D	10.	B

(1 mark each)

B. Problem solving (19 marks)

11. (a) (i) Developing follicle (1)
(ii) The yellow body / Corpus luteum (1)
(b) (i) The rising concentration of oestrogen causes the uterine lining to thicken. (1)
It also inhibits the FSH secretion by the pituitary gland. (1)
(ii) A high concentration of oestrogen stimulates the pituitary gland to secrete FSH and LH. (1)
The peak of oestrogen level leads to a surge in FSH and LH levels which cause ovulation to occur. (1)
(c) The relatively high level of oestrogen from day 16 to day 21 inhibits the pituitary gland, (1)
causing the pituitary gland to secrete less FSH and LH. (1)
The yellow body degenerates due to a low level of LH. (1)
Degenerated yellow body no longer secretes oestrogen, causing the concentration of oestrogen to fall from day 22 to day 28. (1)

(Total: 10 marks)

12. (a) A high level of levonorgestrel inhibits FSH and LH secretions by the pituitary gland. (1)
The reduced level of FSH cannot stimulate follicle development. (1)
This leads to a low level of oestrogen. (1)
The low level of oestrogen, together with a low level of LH, prevents ovulation from occurring. (1)
(b) Ovulation already occurred /
Fertilization or implantation already occurred /
Diarrhoea, vomiting or stomach upset may reduce the absorption of the pill (1)
(c) Few progesterone receptors are available for progesterone because the receptors are occupied by drug X. (1)
Few progesterone receptors are triggered. (1)
The uterine lining cannot maintain its thickness and breaks down. (1)
The implanted embryo is lost together with the shed uterine lining. (1)

(Total: 9 marks)

© 2019 Aristo Educational Press Ltd.

C. Exam challenge (11 marks)

13. (a) Menstruation (1)
The levels of oestrogen and progesterone decrease before menstruation. (1)
The uterine lining breaks down as a result of low levels of oestrogen and progesterone. (1)

(b) (i) Oestrogen (1)

(ii) Oestrogen (1)
Progesterone (1)

(c) Fertilization occurred in the second menstrual cycle. (1)
The embryo implanted into the uterine lining and started to develop a placenta. (1)
The placenta secreted human chorionic gonadotrophin, (1)
which acts like LH to maintain the yellow body. (1)
The yellow body thus continued to secrete oestrogen and progesterone to maintain the thick uterine lining. (1)

(Total: 11 marks)

© 2019 Aristo Educational Press Ltd.